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October 11, 2026

The Editor-in-Chief
Computers & Security
Elsevier

Dear [MISSING: name of the Editor-in-Chief, or keep "Editor"],

We submit the manuscript *"Bounded-memory streaming redaction of personal data and credentials with a whole-document equivalence guarantee"* for consideration as a full research article in *Computers & Security*.

The problem. Organisations routinely share logs, database exports and incident captures that contain personal data and credentials, and these artefacts are often too large to process in memory. Chunked processing is the obvious remedy, but a value that straddles a chunk boundary can escape detection and be written out partly or wholly in the clear. To our knowledge, no redaction, de-identification or secret-scanning tool states when its chunked output equals its whole-document output.

What the manuscript contributes.

- A streaming redaction engine whose output is proven identical to a whole-document pass under two explicit assumptions (bounded span length and bounded context), with the assumption constants derived from each detector's declared requirements and enforced at run time and by regression tests.
- A multi-process mode whose output equals the sequential output byte for byte, with a seam cross-check that falls back to sequential processing on any disagreement, and a working-memory bound independent of input size.
- An evaluation on held-out test splits against Microsoft Presidio and two purpose-built secret scanners, gitleaks and TruffleHog, with paired bootstrap confidence intervals, a component ablation, a contamination analysis for a benchmark used during development, a peak-memory curve from 8 MiB to 2 GiB, and repeated throughput measurements.

On held-out data, the engine's pattern tiers fully redacted 43.7% of test secrets against 31.1% for gitleaks and 27.0% for TruffleHog, at statistically indistinguishable precision; streaming memory stayed at or below 8.5 MB up to 2 GiB of input; and the parallel engine scaled 3.9-fold on four cores. The manuscript also reports where the approach loses, including low free-text recall without a named-entity model.

Fit with the journal. The work addresses data-loss prevention and the safe handling of sensitive operational data, an applied security problem for practitioners who must share logs and exports. Its contributions are an algorithmic guarantee and a systems design; the core engine uses validated patterns and entropy rules, not machine learning. A statistical named-entity model appears only as an optional tier and as part of the baseline comparison. [MISSING: confirm against the journal's current policy on submissions with an AI/ML component before submitting]

Prior dissemination. [MISSING: Choose one and delete the other.]

(a) An earlier version of this system was presented at FTNCT'09 as "SafeStream-Redactor: Provably-Equivalent Constant-Memory Streaming Redaction of PII and Credentials" [MISSING: publication details]. The present manuscript extends it with new technical content not previously published: the held-out test-split evaluation; the secret-scanner baselines; the component ablation; the contamination analysis; the benign-shape suppression in the entropy tier, which reduced the false-alarm rate on generated non-secrets from 0.794 to 0.010 on the test split; the peak-memory curve to 2 GiB; repeated throughput and parallel-scaling measurements; and Rust–Python backend parity results. [MISSING: state the estimated share of new content]

(b) This manuscript has not been published previously, in whole or in part.

The manuscript is not under consideration elsewhere, and all authors have approved its submission.

Openness and declarations. Source code, split manifests, the evaluation harness and all raw results are publicly available at `https://github.com/MounishSenisetty/SafeStream-Redactor`, and every number in the manuscript is generated from those files. The authors declare no competing interests [MISSING: confirm]. As required by Elsevier's policy, the manuscript declares the use of generative AI in preparing the evaluation scripts and the text, which the authors reviewed and take responsibility for.

Thank you for considering our
manuscript.
Yours sincerely,

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on behalf of all authors
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